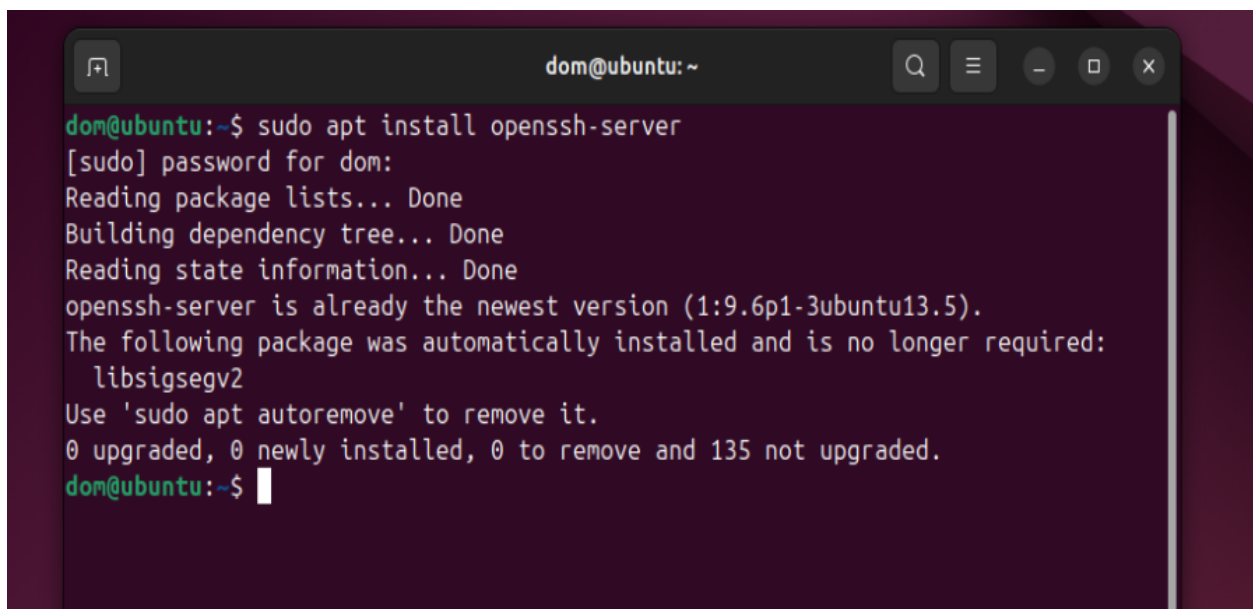


Wazuh Project

This is a home project I did involving the installation and configuration of a Wazuh Server running on Ubuntu. Wazuh is an open-source SIEM tool made to protect organizations against threats in cybersecurity, it is a blue team or defensive tool used in the industry that has and IDS/IPS, event and log analysis tools, vulnerability management, reporting & compliance, extensibility & integrations, scalable architecture, correlation, and threat intelligence. On my Wazuh Server, I am monitoring 2 windows server machines. One is Server 2022; the other is an intentionally vulnerable Server 2016 machine. The Server 2022 machine is a general and properly configured domain controller. The Server 2016 machine is intentionally vulnerable, so I can see what would show up on a vulnerable machine when using Wazuh to monitor it.

A terminal window titled 'dom@ubuntu: ~' with standard window controls. The terminal shows the command 'sudo apt install openssh-server' being executed. The output indicates that the package is already installed and up-to-date, and lists 'libsigsegv2' as a no longer required dependency. The prompt returns to 'dom@ubuntu:~\$' with a cursor.

```
dom@ubuntu:~$ sudo apt install openssh-server
[sudo] password for dom:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.6p1-3ubuntu13.5).
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 135 not upgraded.
dom@ubuntu:~$
```

The first thing I did was install an Ubuntu VM in VMware. Once that was complete, I installed openssh so I can use PuTTY to SSH into the machine to make the configuration easier.

sudo apt install openssh-server installed SSH.

```
dom@ubuntu:~$ sudo systemctl start ssh
dom@ubuntu:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
dom@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
   Active: active (running) since Fri 2025-01-10 19:40:12 EST; 6min ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1248 (sshd)
      Tasks: 1 (limit: 4558)
     Memory: 1.8M (peak: 2.4M swap: 4.0K swap peak: 4.0K)
        CPU: 23ms
     CGroup: /system.slice/ssh.service
            └─1248 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 10 19:40:12 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell >
Jan 10 19:40:12 ubuntu sshd[1248]: Server listening on :: port 22.
Jan 10 19:40:12 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell s>
lines 1-16/16 (END)
dom@ubuntu:~$
```

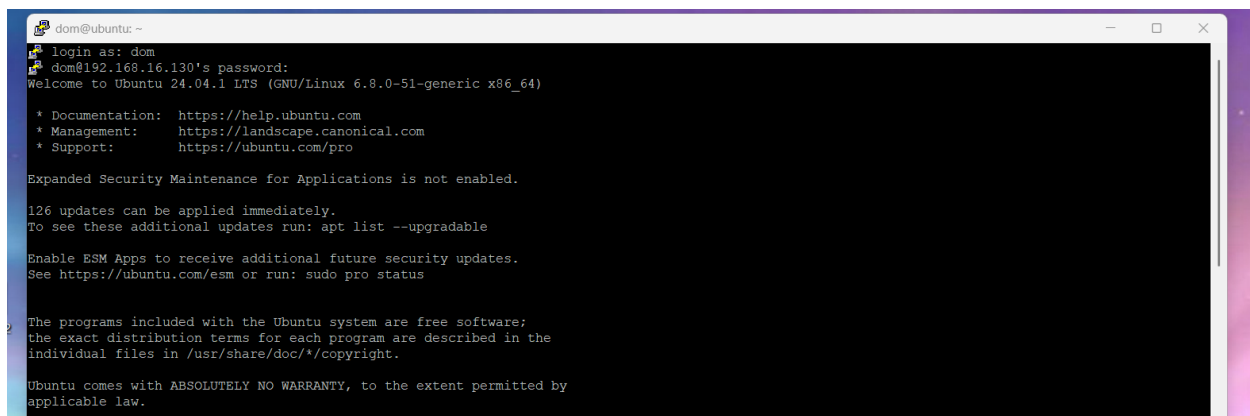
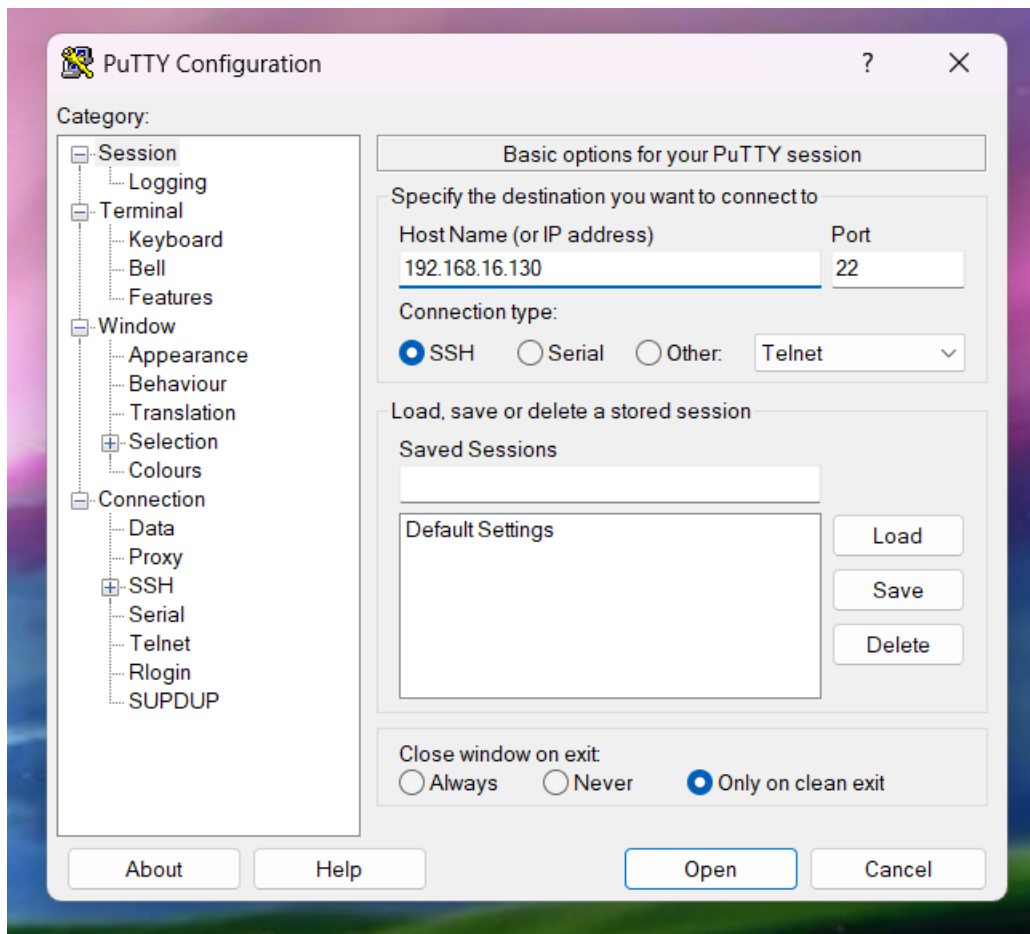
Then I started and enabled SSH, along with checking the status by typing:

sudo systemctl start ssh

sudo systemctl enable ssh

sudo systemctl status ssh

```
dom@ubuntu:~$ sudo apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
The following package was automatically installed and is no longer required:
  libsigsegv2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 135 not upgraded.
dom@ubuntu:~$
```



```

dom@ubuntu:~$ curl -sO https://packages.wazuh.com/4.10/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
[sudo] password for dom:
10/01/2025 08:43:20 INFO: Starting Wazuh installation assistant. Wazuh version: 4.10.0
10/01/2025 08:43:20 INFO: Verbose logging redirected to /var/log/wazuh-install.log
10/01/2025 08:43:23 INFO: --- Dependencies ---
10/01/2025 08:43:23 INFO: Installing gawk.
10/01/2025 08:43:26 INFO: Verifying that your system meets the recommended minimum hardware requirements.
10/01/2025 08:43:26 INFO: Wazuh web interface port will be 443.
10/01/2025 08:43:30 INFO: --- Dependencies ---
10/01/2025 08:43:30 INFO: Installing apt-transport-https.
10/01/2025 08:43:31 INFO: Installing debhelper.
10/01/2025 08:43:41 INFO: Wazuh repository added.
10/01/2025 08:43:41 INFO: --- Configuration files ---
10/01/2025 08:43:41 INFO: Generating configuration files.
10/01/2025 08:43:41 INFO: Generating the root certificate.
10/01/2025 08:43:41 INFO: Generating Admin certificates.
10/01/2025 08:43:41 INFO: Generating Wazuh indexer certificates.
10/01/2025 08:43:42 INFO: Generating Filebeat certificates.
10/01/2025 08:43:42 INFO: Generating Wazuh dashboard certificates.
10/01/2025 08:43:42 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
10/01/2025 08:43:42 INFO: --- Wazuh indexer ---
10/01/2025 08:43:42 INFO: Starting Wazuh indexer installation.
10/01/2025 08:44:16 INFO: Wazuh indexer installation finished.

```

Then I established my SSH session in PuTTY. I also needed to install curl on Ubuntu.

`sudo apt install curl` installed it.

Once curl was installed, I entered the command `curl -sO`

<https://packages.wazuh.com/4.10/wazuh-install.sh> && `sudo bash ./wazuh-install.sh -a` which began the installation of Wazuh. The version I installed was Wazuh 4.10.

```

10/01/2025 08:43:42 INFO: --- Wazuh indexer ---
10/01/2025 08:43:42 INFO: Starting Wazuh indexer installation.
10/01/2025 08:44:16 INFO: Wazuh indexer installation finished.
10/01/2025 08:44:16 INFO: Wazuh indexer post-install configuration finished.
10/01/2025 08:44:16 INFO: Starting service wazuh-indexer.
10/01/2025 08:44:25 INFO: wazuh-indexer service started.
10/01/2025 08:44:25 INFO: Initializing Wazuh indexer cluster security settings.
10/01/2025 08:44:27 INFO: Wazuh indexer cluster security configuration initialized.
10/01/2025 08:44:27 INFO: Wazuh indexer cluster initialized.
10/01/2025 08:44:27 INFO: --- Wazuh server ---
10/01/2025 08:44:27 INFO: Starting the Wazuh manager installation.
10/01/2025 08:45:28 INFO: Wazuh manager installation finished.
10/01/2025 08:45:29 INFO: Wazuh manager vulnerability detection configuration finished.
10/01/2025 08:45:29 INFO: Starting service wazuh-manager.
10/01/2025 08:45:41 INFO: wazuh-manager service started.
10/01/2025 08:45:41 INFO: Starting Filebeat installation.
10/01/2025 08:45:46 INFO: Filebeat installation finished.
10/01/2025 08:45:47 INFO: Filebeat post-install configuration finished.
10/01/2025 08:45:47 INFO: Starting service filebeat.
10/01/2025 08:45:48 INFO: filebeat service started.
10/01/2025 08:45:48 INFO: --- Wazuh dashboard ---
10/01/2025 08:45:48 INFO: Starting Wazuh dashboard installation.
10/01/2025 08:46:23 INFO: Wazuh dashboard installation finished.
10/01/2025 08:46:23 INFO: Wazuh dashboard post-install configuration finished.
10/01/2025 08:46:23 INFO: Starting service wazuh-dashboard.
10/01/2025 08:46:24 INFO: wazuh-dashboard service started.
10/01/2025 08:46:25 INFO: Updating the internal users.
10/01/2025 08:46:28 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
10/01/2025 08:46:36 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
10/01/2025 08:47:02 INFO: Initializing Wazuh dashboard web application.
10/01/2025 08:47:03 INFO: Wazuh dashboard web application initialized.
10/01/2025 08:47:03 INFO: --- Summary ---
10/01/2025 08:47:03 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: KBwfs1bn0j52EpGDvymgglgc.0WZ*L
10/01/2025 08:47:03 INFO: --- Dependencies ---
10/01/2025 08:47:03 INFO: Removing gawk.
10/01/2025 08:47:05 INFO: Installation finished.

```

I made sure the installation was successful. At the bottom, it states my login credentials, as well as the indication that the server is running on port 443 or HTTPS using a self-signed certificate.

```

dom@ubuntu:~$ sudo /var/ossec/bin/agent_control -l
[sudo] password for dom:

Wazuh agent_control. List of available agents:
  ID: 000, Name: ubuntu (server), IP: 127.0.0.1, Active/Local

List of agentless devices:

```

Before I logged in, I ran the command **sudo /var/ossec/bin/agent_control -l** that showed the available agents configured in the Wazuh environment. Only the Ubuntu machine was present when I issued this command at first.

```
dom@ubuntu:~$ sudo /var/ossec/bin/manage_agents

*****
* Wazuh v4.10.0 Agent manager.                *
* The following options are available:         *
*****
  (A)dd an agent (A).
  (E)xtract key for an agent (E).
  (L)ist already added agents (L).
  (R)emove an agent (R).
  (Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
  Please provide the following:
    * A name for the new agent: server2016
    * The IP Address of the new agent: 192.168.16.166
Confirm adding it?(y/n): y
Agent added with ID 001.

*****
* Wazuh v4.10.0 Agent manager.                *
* The following options are available:         *
*****
  (A)dd an agent (A).
  (E)xtract key for an agent (E).
  (L)ist already added agents (L).
  (R)emove an agent (R).
  (Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
  Please provide the following:
    * A name for the new agent: server2022
    * The IP Address of the new agent: 192.168.16.135
Confirm adding it?(y/n): y
Agent added with ID 002.

*****
```

Then I ran the command **sudo /var/ossec/bin/manage_agents** which opened the agent manager for Wazuh. In this, I configured the 2 Windows machines that I used for monitoring within Wazuh. For Server 2016, the name I set was **server2016**, and the IP was **192.168.16.166**. The ID was 001.

For Server 2022, the name I set was **server2022**, and the IP was **192.168.16.135**. The ID was 002.

```
*****
* Wazuh v4.10.0 Agent manager. *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: E

Available agents:
ID: 001, Name: server2016, IP: 192.168.16.166
ID: 002, Name: server2022, IP: 192.168.16.135
Provide the ID of the agent to extract the key (or '\q' to quit): 001

Agent key information for '001' is:
MDAxIHn1cnZlcjIwMTYgMTkyLjE2OC4xNi4xNjYgN2VhNzdiYzU3Yjg3MTc0ZmU4ZmM4MmZlMWM5MWIwODdkZTBiMWQzNDAYNWJmNDNjODY1YmVmYTkyMTY2MTJkOA==

** Press ENTER to return to the main menu.

*****
* Wazuh v4.10.0 Agent manager. *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: E

Available agents:
ID: 001, Name: server2016, IP: 192.168.16.166
ID: 002, Name: server2022, IP: 192.168.16.135
Provide the ID of the agent to extract the key (or '\q' to quit): 002

Agent key information for '002' is:
MDAYIHn1cnZlcjIwMjIwMTYgMTkyLjE2OC4xNi4xMzUgYzF1ZWl4MGQxNDMxODM1NTE4ZTg4ODJkYzRhZTEyY2E3NWY2YWE5NWVlZGFlZWVhZTQ3N2JkOGI4NDVlZWYwNA==

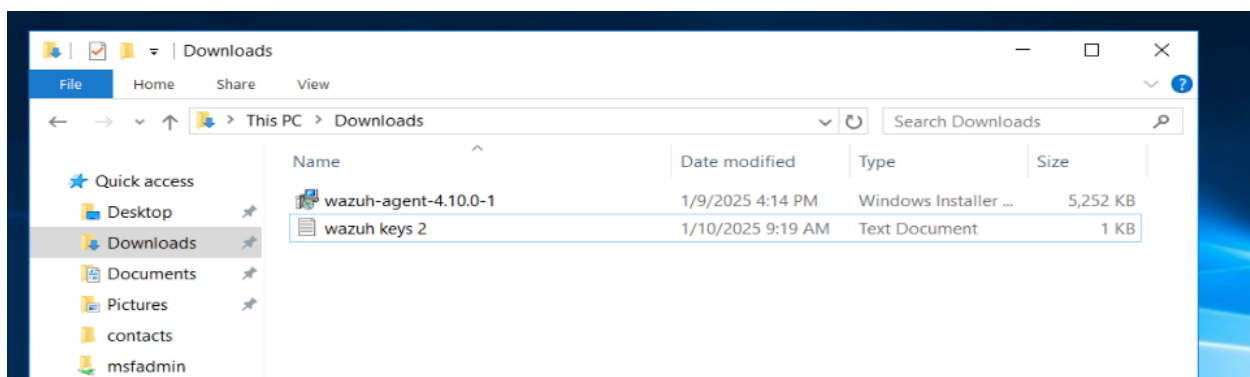
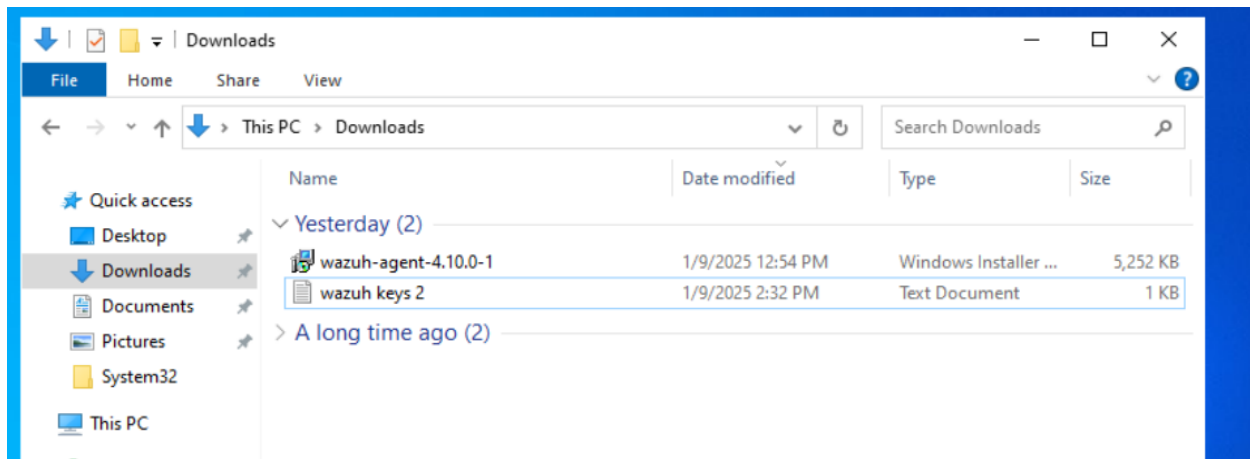
** Press ENTER to return to the main menu.
```

Then back in the agent, I needed to extract keys for each machine for the agents to work properly when configuring the agents on each windows machine. To do this, I chose option E, and entered each machine’s respected ID, which generated the key for each machine.

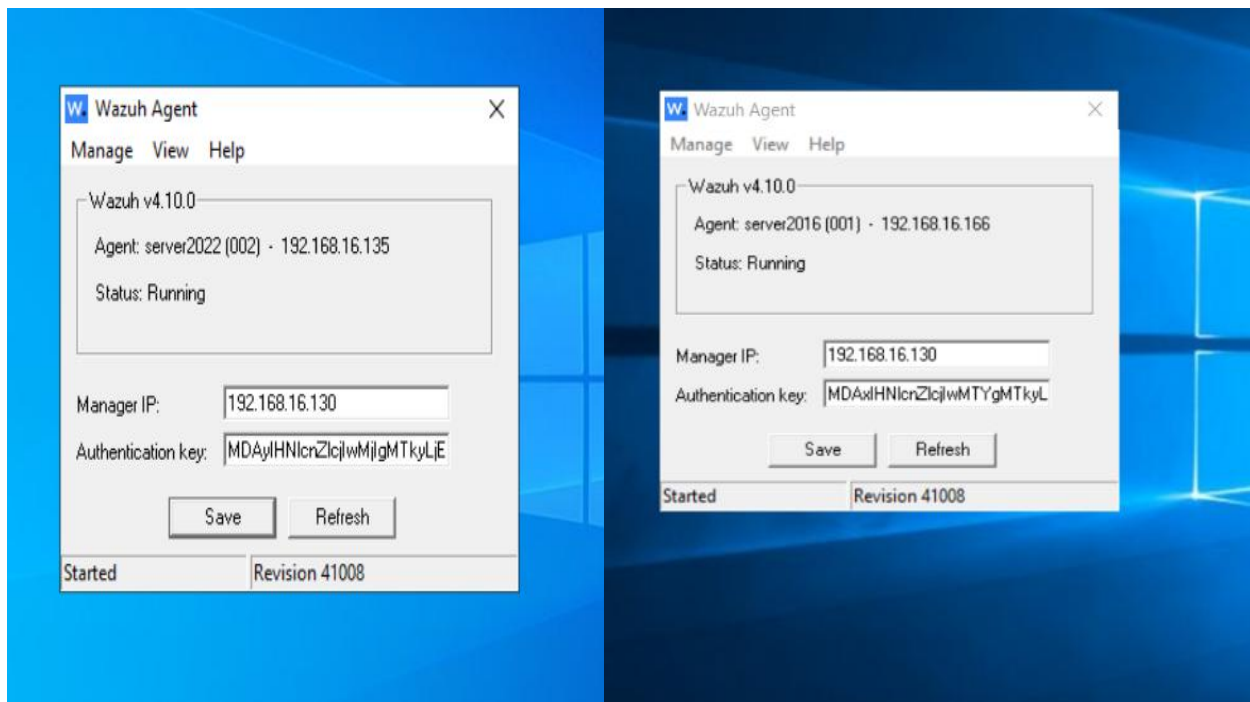
```
server 2016
MDAxIHn1cnZlcjIwMTYgMTkyLjE2OC4xNi4xNjYgN2VhNzdiYzU3Yjg3MTc0ZmU4ZmM4MmZlMWM5MWIwODdkZTBiMWQzNDAYNWJmNDNjODY1YmVmYTkyMTY2MTJkOA==

server 2022
MDAYIHn1cnZlcjIwMjIwMTYgMTkyLjE2OC4xNi4xMzUgYzF1ZWl4MGQxNDMxODM1NTE4ZTg4ODJkYzRhZTEyY2E3NWY2YWE5NWVlZGFlZWVhZTQ3N2JkOGI4NDVlZWYwNA==
```

I then copied those keys and pasted them into a text document on my host machine.

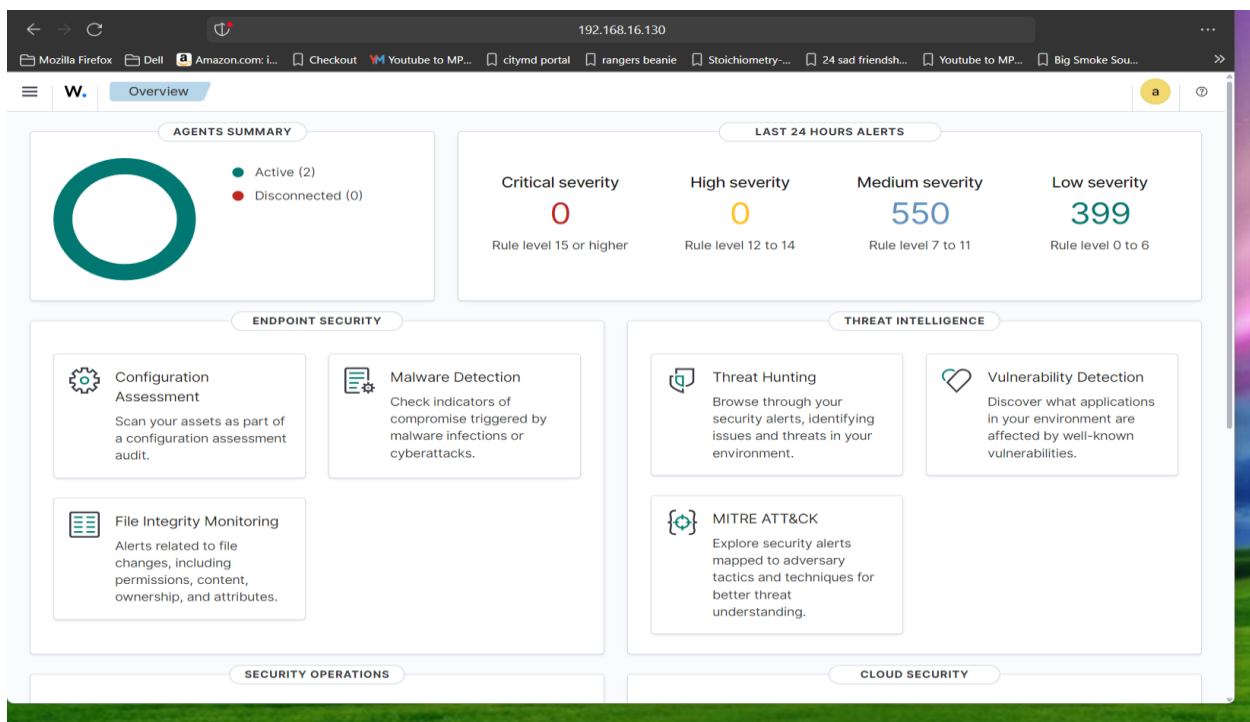


I uploaded the keys file to my Dropbox account. On Server 2022 and 2016, I went onto Dropbox and downloaded the file, and I also downloaded the agent setup file to configure Wazuh on each Windows machine.



Then on each Windows machine, I installed the Wazuh agent, and after the installation, I opened the agent. For the manager IP, I typed the IP of the Ubuntu machine I was using where the Wazuh server is configured being **192.168.16.130**.

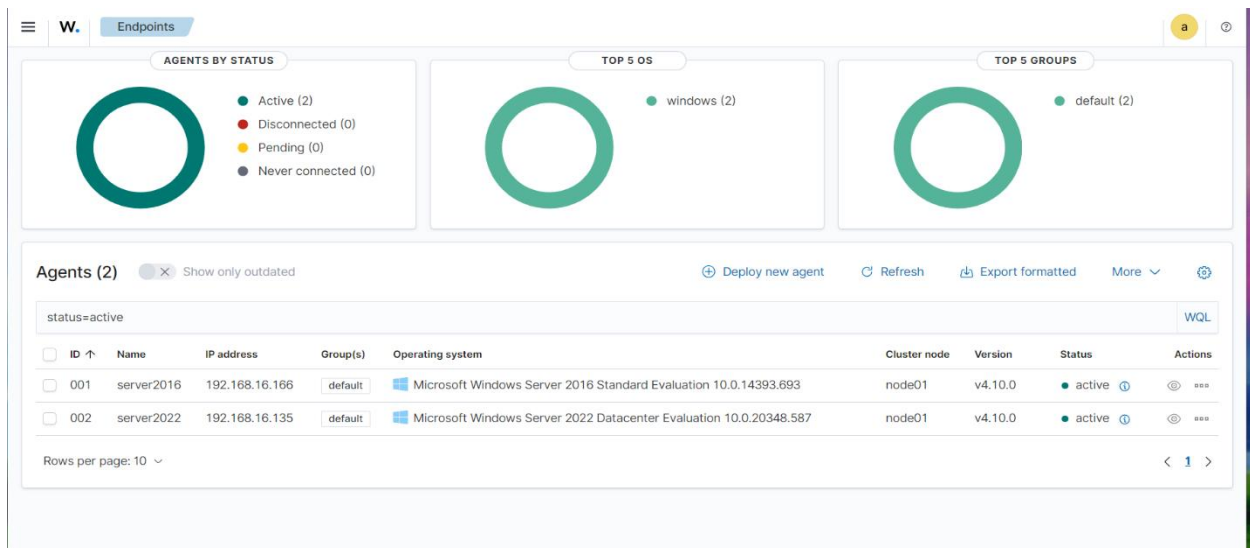
For the authentication key, I pasted the key for each respected machine. I then clicked the save button.



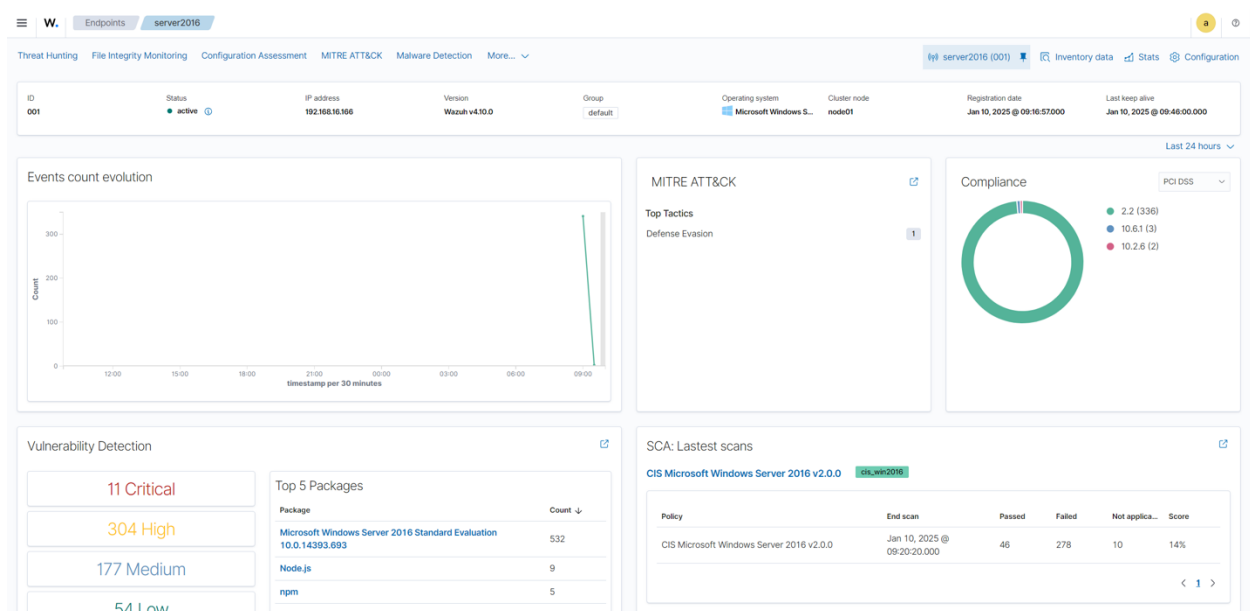
Then on my host machine, I typed in **https://<ubuntu ip here>/**

It initially gave me a security warning because of the self-signed certificate use, but I advanced and went to the website anyway since it is safe.

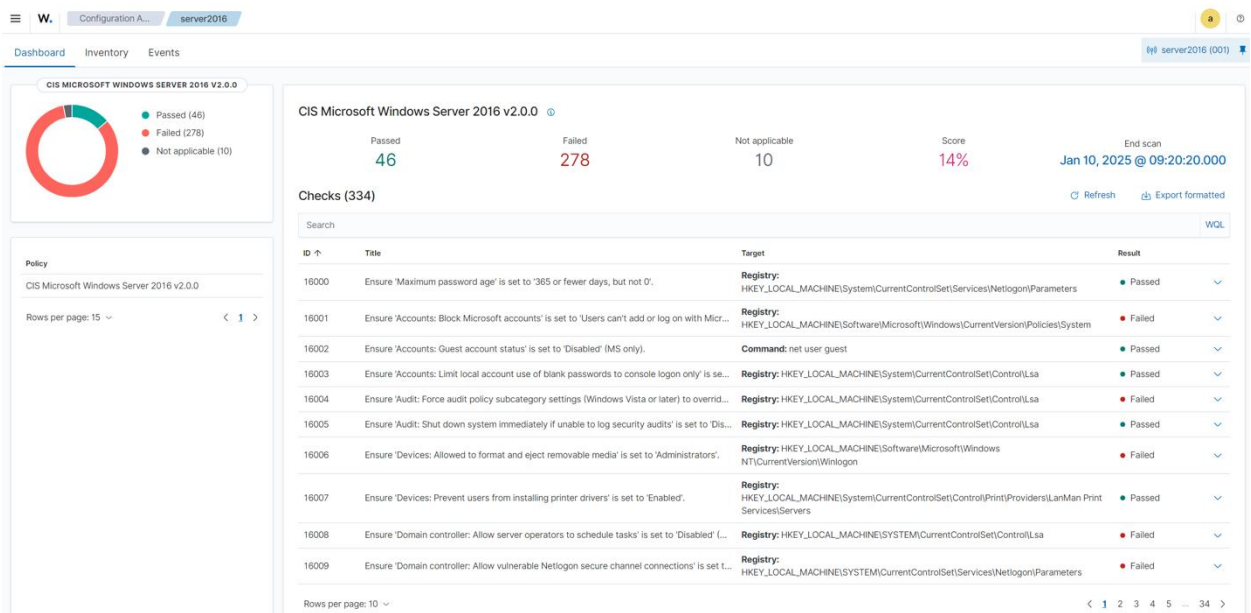
This is the homepage of Wazuh. Where 'agents summary' is, it states there are 2 active machines. Those 2 active machines are the 2 Windows machines I configured.



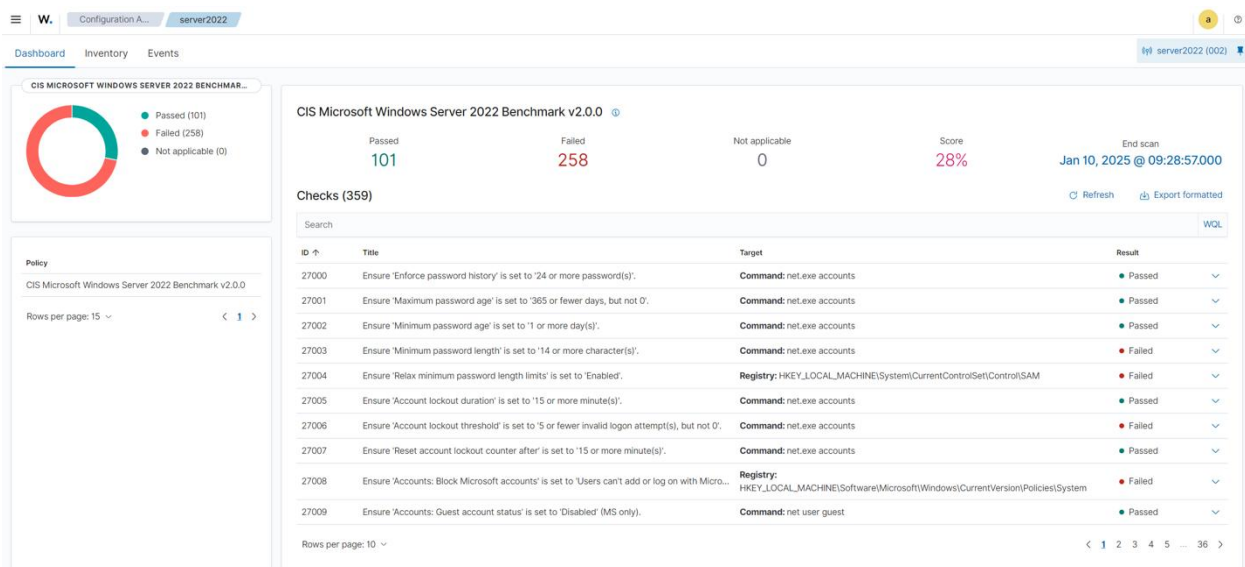
This is an endpoints overview in Wazuh. This shows both active agents, being both Windows machines in my case. In the endpoints overview, the status is shown, being active, the operating systems, being Windows Server 2022 and Windows Server 2016, the group it belongs to is the default one, their respected IP addresses, cluster node, and version.



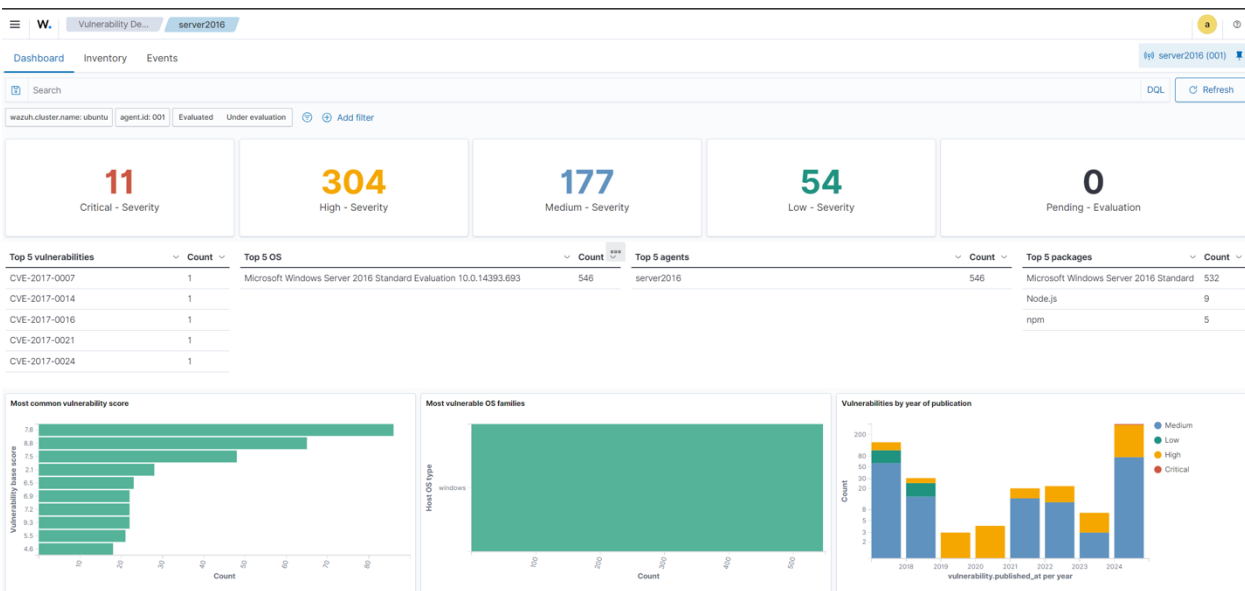
This screenshot is detailed monitoring of the Server 2016 machine. This highlights the monitoring of this agent. What is shown is the IP address, status and operating system. The top MITRE ATT&CK tactics are also displayed focusing on evasion of defense, as well as the compliance report for the PCI DSS standards, and metrics for vulnerability detection.



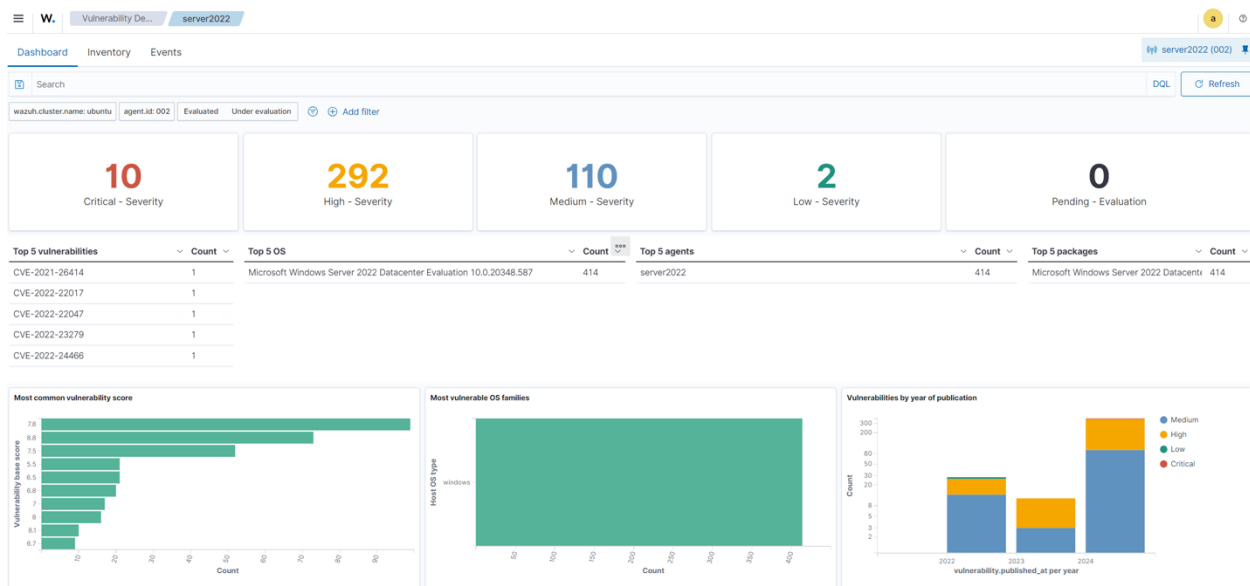
This screenshot is displaying the CIS or Center for Internet Security benchmark test for the Server 2016 machine. This is displaying the total number of checks performed, including counts of passed, failed and checks that aren't applicable. It also displays the details of the failed checks, like password policy ensurement and the configuration of accounts meet the best practices.



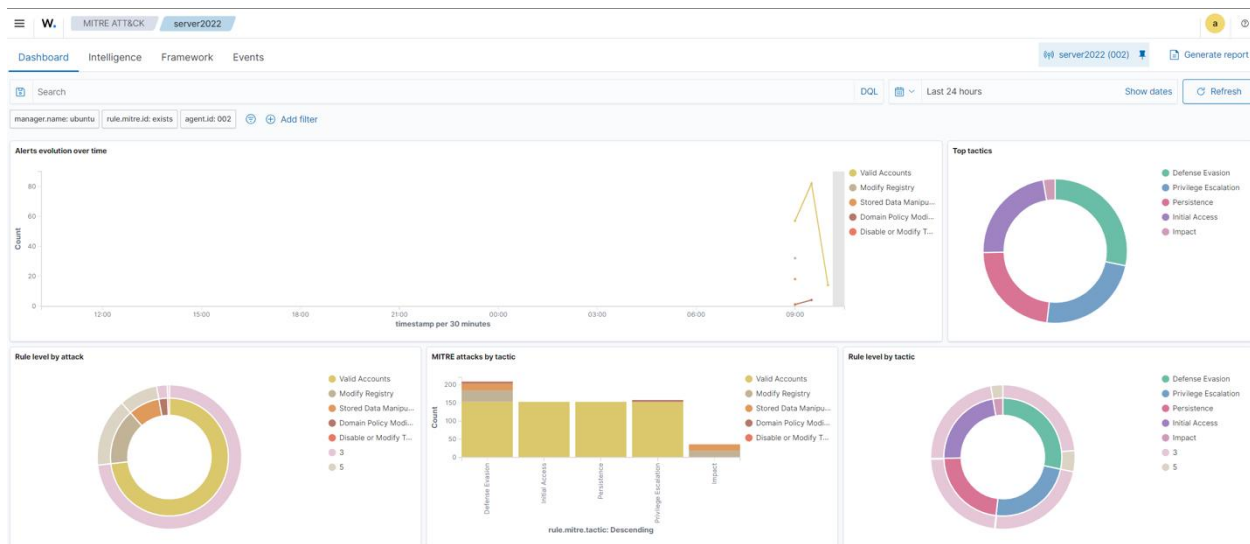
This is like the previous screenshot; this is displaying the CIS for the Server 2022 machine. This one is not as bad the Server 2016 machine since this machine is not intentionally vulnerable. This one includes failed and passed checks related to password policies, access for local users, as well as account lockout thresholds.

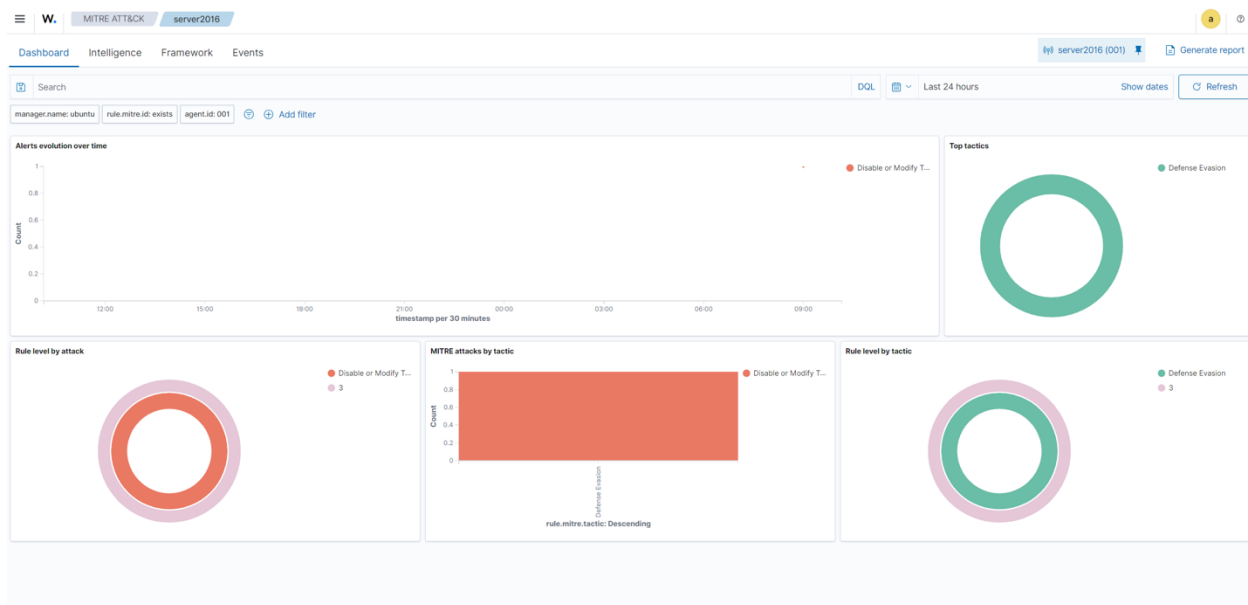


This screenshot is showing vulnerability scan results for the Server 2016 machine. The key points covered in this are the severity levels of each vulnerability, ranging from critical, high, medium and low, the top vulnerabilities that were detected, as well as the vulnerability breakdown by the operating system family and year of publication.



This is like the previous screenshot, but this is for Server 2022. The same concepts are highlighted; the severity levels and counts of the vulnerabilities, the CVEs involved with detected vulnerability, and the breakdown of each by OS family.





These 2 screenshots highlight the MITRE ATT&CK dashboard for both Server 2022 and Server 2016.

For Server 2022, the features included are the evolution of alerts overtime, attacks categorized by tactics like initial access, privilege escalation and persistence, as well as rule levels by attack and tactic-based visualizations.

For Server 2016, the main focuses for this machine are a single detected attack being defense evasion. This machine also displaying for this dashboard, the related rules and attack mappings for this one tactic.

Works Cited

<https://medium.com/@DaoudaD/wazuh-part-1-presentation-implementation-5806297b9824>

<https://www.youtube.com/watch?v=u2pS0Zuzbmc&list=LL&index=1>